Ty Bohan

Senior AI Security Engineer | Adversarial AI & Agentic Red Teaming

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PROFILE

Senior AI security engineer with 10+ years in offensive security specializing in adversarial evaluation of generative AI systems, autonomous agents, and large-scale ML pipelines.

At Amazon (Stores Security), designed and built an agentic AI red-teaming platform from the ground up using LangGraph and custom orchestration frameworks that became the org's standard for continuous adversarial verification of LLMs, AI workflows, and agentic systems in retail/consumer experiences—replacing external vendors and delivering multimillion-dollar operational savings.

Expert in uncovering systemic gaps between AI security claims and real-world behavior through deep experimentation, automation, and adversarial modeling. Proven leader translating complex AI risk into actionable strategy across engineering and executive stakeholders.

U.S. Coast Guard veteran, CVE author, and speaker at OffensiveCon and ShmooCon.

EMPLOYMENT HISTORY

Jan 2025 — Remote

Offensive Security Engineer, Amazon

- Architected and operationalized an agentic AI red-teaming platform using LangGraph and custom orchestration to continuously stress-test LLM guardrails and multi-agent systems, leading a 4-engineer team to establish it as Amazon Stores' 2026 AI security standard.
- Delivered $50M in operational savings and 35% higher testing efficiency through GenAI-augmented automation within the red-teaming platform; institutionalized adversarial intelligence and shifted specialized teams from manual efforts to scalable discovery of high-criticality vulnerabilities.
- Developed an AI-driven discovery pipeline analyzing adversarial prompt behavior across 150+ assessments, exposing systemic guardrail failures, saving 300+ manual hours annually, and building a data foundation for cross-org AI security research.
- Shaped AI security strategy via gap analysis of 7+ industry solutions; identified critical multi-turn/inference gaps, influencing roadmaps across internal security tool teams.
- Designed and deployed automated governance system (AWS Lambda/Python) for third-party component assessments; eliminated systemic missed findings in 3P integrations, achieving zero undeclared incidents post-launch.

Mar 2023 — Dec 2024 — Remote

Senior Security Engineer & Vulnerability Researcher, Cisco Meraki

- Engineered a runtime introspection framework for Ruby on Rails APIs, enabling continuous schema monitoring and automated regression detection; institutionalized this as a core security guardrail that eliminated 90% of IDOR and authorization vulnerabilities across 100+ endpoints.
- Architected cloud-native Secure SDLC infrastructure (Python/IaC) leveraging Semgrep for custom detection engineering; institutionalized a feedback loop where offensive findings were converted into automated CI/CD guardrails, preventing vulnerability regressions across 100+ global repositories.
- Pioneered firm-wide engineering governance initiative by designing graph-based asset catalog (open-source frameworks + YAML-as-code); orchestrated legacy data migration and established core data structures mapping security posture across the Meraki ecosystem.

Mar 2022 — Nov 2022 — Remote

Senior Security Engineer, Stripe

- Conducted in-depth source code reviews and security assessments across application and cloud infrastructure layers; identified high-risk areas and drove implementation of targeted controls.
- Designed and validated technical controls for PCI DSS compliance; coordinated with external auditors and led compliance interviews for payment processing infrastructure.
- Contributed to development and enforcement of security policies and standards aligned with PCI DSS and SOC 2 requirements.

- Influenced product roadmap and security investments by communicating offensive findings, exploitability, and compliance gaps to senior leadership.

Mar 2019 — Mar 2022

Security Engineer II (Hardware Security), Amazon

Remote

- Led security architecture reviews, threat modeling, and risk assessments for Amazon-branded hardware products; focused on secure boot, HSMs, and cryptographic key management.
- Designed and implemented security controls in partnership with hardware, firmware, and platform teams; mitigated risks from device compromise, firmware attacks, and cryptographic key exposure.
- Oversaw deployment of hardware-backed key storage solutions to protect cryptographic material against unauthorized access and extraction.
- Conducted adversary-driven penetration testing on embedded systems and IoT devices; identified attack paths across firmware, hardware, and cloud backends.
- Developed automated testing tools and scripts to enhance efficiency, consistency, and coverage of hardware security assessments.

SECURITY RESEARCH & CONTRIBUTIONS

- **Vulnerability Research:** Discovered and responsibly disclosed multiple CVEs across major vendors, including critical iOS user-space remote code execution vulnerabilities (CVE-2016-4631, CVE-2016-4637), enabling reliable code execution within app sandbox constraints.
- **Conference Speaking**: Presented original offensive security research at ShmooCon (2016), SummerCon (2016), Sec-T Stockholm (2016), and OffensiveCon Berlin (2019).
- **Open-Source Contributions:** Developed MacHeap (advanced LLDB Python framework for macOS heap introspection) and an AI-powered interactive framework showcasing security automation and GenAI integration.

TECHNICAL SKILLS

- **Languages**: Python, C/C++, Go, Assembly, Bash/Shell, SQL
- **Cloud & Infrastructure**: AWS (EC2, S3, RDS, Lambda, API Gateway), Docker, Kubernetes, Terraform/IaC, CI/CD, Linux/Unix, Windows
- **Offensive Security**: Penetration Testing, Red Teaming, Vulnerability Research, Exploit Development, Mobile/IoT/Hardware Security, Dynamic/Static Analysis (SAST/DAST)
- **Frameworks & Standards**: Metasploit, Burp Suite Pro, Kali Linux, Custom Offensive Frameworks, OSINT, Threat Modeling, GenAI Automation, MITRE ATT&CK, MITRE ATLAS, OWASP Top 10 (Web & LLM), CVSS, PCI DSS, SOC 2, HIPAA, GDPR
- **AI Security Expertise:** LLM adversarial testing • AI lifecycle risk analysis • Multi-agent security • AI red-team automation (LangGraph) • AI-driven vulnerability discovery

EDUCATION

New York University, Coursework in Computer Science